Practice plan: node setup and operation drills

Twelve drills to repeat until each one is boring, from a throwaway node to the real one. Use with the build, auto-start, security and withdrawal tutorials.

Three places to practise

Place	Money	Use it for	On this Mac
Regtest (Polar)	Fake, unlimited, you mine blocks	Breaking things: force closes, restores, wrong keys	Polar.app is installed. Also the bootcamp bitcoind (port 18443)
Signet	Free test coins from a faucet	Real network behaviour: peers, routing, waiting for blocks	Any LND with bitcoin.signet=true and its own lnddir
Mainnet	Real	Only drills you have already done on regtest and signet	The Altradits node. Small amounts

Rule for practice machines: each practice node gets its own folder (for example `--lnddir=$HOME/.lnd-practice`) and its own ports, and never the mainnet seed. On another computer, follow the same drills from Drill 1.

The drills

Drill 1. Build a node from zero

Goal: go from nothing to a synced node **Where:** Polar, then signet **Repeat:** 5 times

```
Polar: New Network -> 1 bitcoind + 2 LND nodes -> Start
Signet: lnd --lnddir=$HOME/.lnd-practice --bitcoin.signet --bitcoin.node=neutrino
```

Done when: getinfo shows synced_to_chain true.

Drill 2. Create a wallet and restore it

Goal: trust your seed **Where:** Regtest only **Repeat:** 3 times

```
lncli create              # write the 24 words on paper
# stop LND, delete the practice folder, start again
lncli create              # choose y, type the seed back in
```

Done when: The restored wallet shows the same on-chain address history and balance.

Drill 3. Read-only tour

Goal: know every number on the dashboard **Where:** Mainnet (safe: read-only) **Repeat:** weekly

```
ln-ro getinfo; ln-ro listchannels; ln-ro channelbalance
ln-ro walletbalance; ln-ro listpayments; ln-ro listinvoices
```

Done when: You can explain each figure on the Money tab from the command output.

Drill 4. Receive a payment

Goal: make and track an invoice **Where:** Regtest, then mainnet with 1,000 sats **Repeat:** 5 times

```
lncli addinvoice --amt 1000 --memo practice
# pay it from the other node or a wallet
ln-ro lookupinvoice <r_hash>
```

Done when: The invoice state is SETTLED.

Drill 5. Withdraw safely

Goal: send without mistakes **Where:** Regtest, then mainnet small **Repeat:** 5 times

```
ln-ro decodepayreq <invoice>
lncli payinvoice --fee_limit 200 --timeout 5m <invoice>     # type yes in full
```

Done when: SUCCEEDED, and your side of the channel went down by amount plus fee.

Drill 6. Handle a failed payment

Goal: stay calm when routing fails **Where:** Regtest **Repeat:** 3 times

```
# pay more than the channel can send, or to a node with no route
ln-ro listpayments --include_incomplete
```

Done when: You can read the failure reason and show that no sats left.

Drill 7. Keys and permissions

Goal: see what each key can't do **Where:** Regtest **Repeat:** 2 times

```
lncli bakemacaroon --save_to inv.mac invoices:read invoices:write
lncli --macaroonpath=inv.mac payinvoice <invoice>     # must fail: permission denied
```

Done when: The invoice key makes invoices and is refused when it tries to pay.

Drill 8. Open and close a channel

Goal: the full channel life **Where:** Regtest **Repeat:** 3 times

```
lncli openchannel --node_key <peer> --local_amt 1000000
# mine 6 blocks in Polar
lncli closechannel --funding_txid <txid> --output_index <n>
```

Done when: The channel appears, works, closes, and the sats return on-chain.

Drill 9. Force close

Goal: see why it's a last resort **Where:** Regtest **Repeat:** 2 times

```
# stop the peer, then:
lncli closechannel --force --funding_txid <txid> --output_index <n>
ln-ro pendingchannels
```

Done when: You see blocks_til_maturity counting down before your funds are spendable.

Drill 10. Disaster recovery

Goal: survive a dead machine **Where:** Regtest **Repeat:** 2 times

```
cp .../channel.backup ~/practice-backup
# delete the node folder
lncli create   # y, seed, then: lncli restorechanbackup --multi_file ~/practice-backup
```

Done when: Channels close and the funds come back to the restored wallet.

Drill 11. Restart and auto-start

Goal: know the node comes back **Where:** Mainnet **Repeat:** after each update

```
launchctl kickstart -k gui/$(id -u)/com.altradits.lnd
launchctl list | grep altradits
```

Done when: After about a minute: synced, channel active, and you deleted the new admin key.

Drill 12. Teach it back

Goal: prove you understand **Where:** Paper or a friend **Repeat:** monthly

```
Explain without notes: keys, reserve, inbound, failed vs succeeded,
why two copies of a node lose money.
```

Done when: Someone else can follow your explanation and repeat Drill 5.

Practice log

Keep one line per session. Mistakes are the useful part: write what happened and what you would do differently.

Date	Drill	Place	What went wrong	What I learned